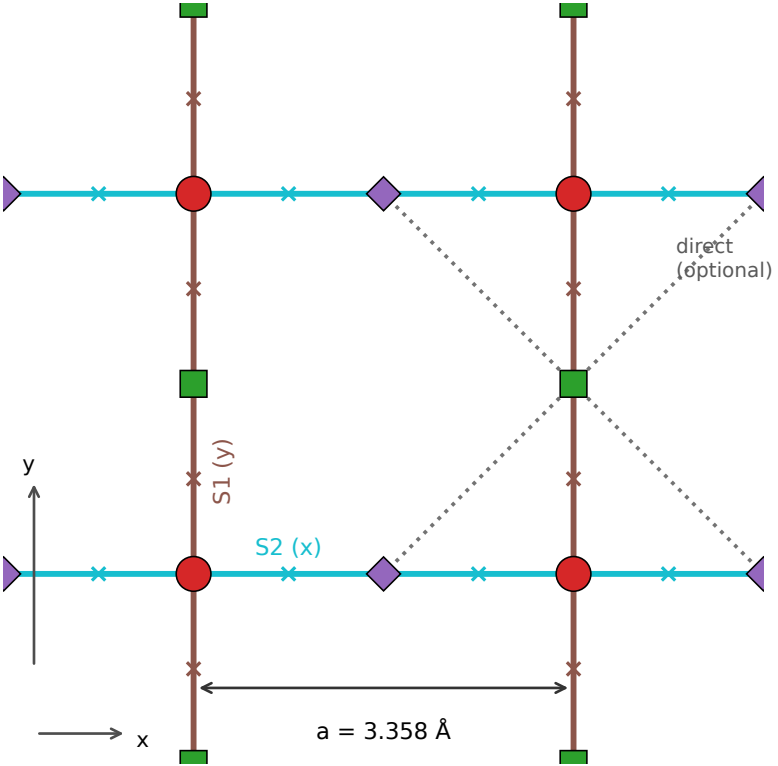
Site network for H on Nb-terminated NbP (one T, B1, B2 per cell; 1 ML barriers)



T (atop Nb)
B1 (Nb–Nb bridge, DFT r0)
B2 (Nb–Nb bridge)
B1 ↔ T via S1 (along y)
T ↔ B2 via S2 (along x)
direct B1 ↔ B2 (sensitivity only)
surface unit cell

Energies vs B1 (eV), M1 / M0

B1	0.000 /	0.000
B2	0.008 /	0.010
T	0.187 /	0.238
S1 (B1↔T)	0.253 /	0.296
S2 (T↔B2)	0.231 /	0.279
direct, grid ≤	0.379 /	0.423

hop length a/2 = 1.679 Å (direct: 2.37 Å)

MLIP-based harmonic-TST estimates; lattice model with 1 ML collective barriers; not DFT-validated; valid only above ~150–200 K